

Physics A (H156, H556)

Chapter 18 Gravitational fields

KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

Candidate forename		Candidate surname	
-----------------------	--	----------------------	--

Centre number						Candidate number				
---------------	--	--	--	--	--	------------------	--	--	--	--

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is **142**.
- The total number of marks may take into account some 'either/or' question choices.

- 1 Two spheres each of radius 10 m and mass 1600 kg exert a gravitational force of 1.7×10^{-8} N on each other.

What is the shortest distance between the surfaces of the spheres?

- A 80 m
- B 100 m
- C 120 m
- D 9900 m

Your answer

[1]

- 2 At the surface of a planet with radius r the magnitude of the gravitational field strength is g .

What is the escape velocity from the surface of the planet?

- A $\sqrt{rg}$
- B $\sqrt{2g}$
- C $\sqrt{2rg}$
- D $2rg$

Your answer

[1]

- 3 Two identical spheres, each of mass 8700 kg, have a space of 3.6 m between their centres.

What is the magnitude of the gravitational force they exert on each other?

- A 2.0×10^{-3} N
- B 3.9×10^{-4} N
- C 7.5×10^{-4} N
- D 4.5×10^{-8} N

Your answer

[1]

- 4 A satellite is in geostationary orbit 36 000 km above the Earth's surface. The Earth has a radius of 6400 km.

At what speed is the satellite moving relative to the centre of the Earth?

- A 0 m s⁻¹
- B 490 m s⁻¹
- C 2.6 km s⁻¹
- D 3.1 km s⁻¹

Your answer

[1]

- 5 Which pair of quantities do **not** have the same, or equivalent, units?

- A acceleration, gravitational field strength
- B angular frequency, angular velocity
- C gravitational potential, kinetic energy
- D impulse, momentum

Your answer

[1]

- 6 Kepler-90 is a star with several planets orbiting it.

The two outermost planets are Kepler-90g and Kepler-90h.

Kepler-90g has an orbital period of 210 days and is 0.71 AU from the centre of Kepler-90.

Kepler-90h is 1.01 AU from the centre of Kepler-90.

Kepler's third law of planetary motion can be applied to the planets of Kepler-90.

What is the orbital period of Kepler-90h?

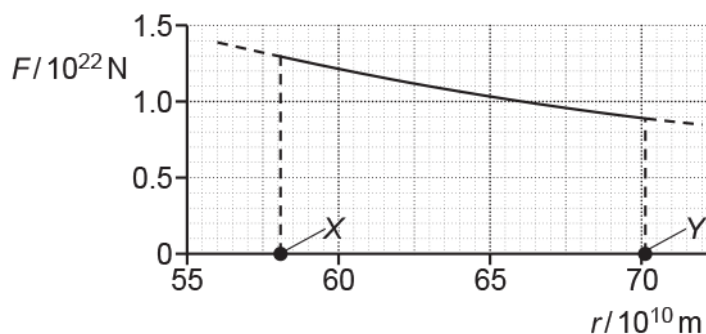
- A 50 days
- B 299 days
- C 356 days
- D 4350 days

Your answer

[1]

- 7 The planet Mercury has a highly elliptical orbit around the Sun.

The gravitational force F acting on Mercury due to the Sun varies with its distance r from the centre of the Sun. The graph of F against r for Mercury in its orbit is shown below.



Mercury is closest to the Sun when $r = X$ and furthest when $r = Y$.

What does the **area** under the graph between the distances X and Y represent?

- A The centripetal force acting on Mercury.
- B The change in the gravitational potential energy of Mercury.
- C The impulse of the force acting on Mercury.
- D The kinetic energy of Mercury.

Your answer

[1]

- 8 The gravitational force between two point-mass objects X and Y is F_1 .

The mass of X increases and the distance between X and Y is halved.

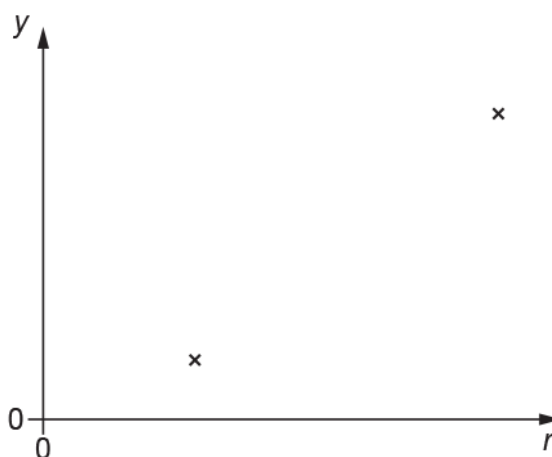
Which statement about the new gravitational force F_2 between these two objects is correct?

- A $0 < F_2 < 0.25F_1$
- B $F_2 > 4F_1$
- C $F_2 = F_1$
- D $2F_1 < F_2 < 4F_1$

Your answer

[1]

- 9 A student has collected some data on the Solar System.
The student plots a graph, but only two data points are shown below.



The distance from the centre of the Sun is r .

Which quantity y is represented on the vertical axis?

- A Speed of a planet.
- B Period of a planet.
- C Gravitational potential of the Sun.
- D Gravitational field strength of the Sun.

Your answer

[1]

10 The Earth is surrounded by a gravitational field.

Which of the following statements is/are correct about the gravitational field lines near the surface of the Earth.

- 1 They are parallel.
- 2 They show the direction of the force on a small mass.
- 3 They are equally spaced.

- A Only 1
- B Only 1 and 2
- C Only 2 and 3
- D 1, 2 and 3

Your answer

[1]

11 Earth has a mass of 6.0×10^{24} kg and a radius of 6400 km.

A satellite of mass 320 kg is lifted from the Earth's surface to an orbit 1200 km above its surface.

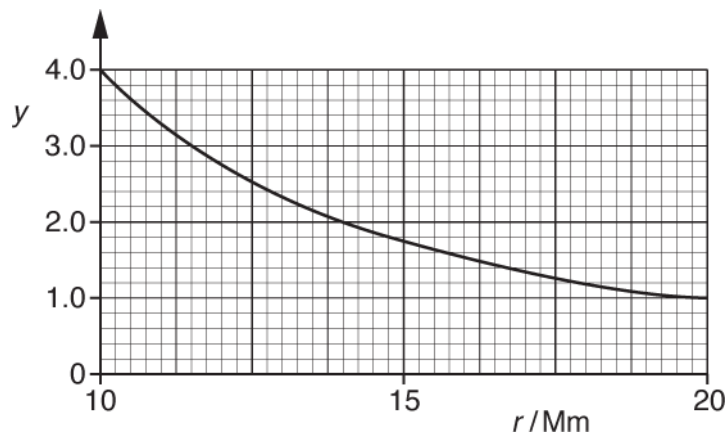
What is the change in the gravitational potential energy of the satellite?

- A 9.1×10^2 J
- B 9.9×10^6 J
- C 3.2×10^9 J
- D 3.8×10^9 J

Your answer

[1]

12 A graph of y against distance r from the centre of a planet is shown below.



The graph shows that y is inversely proportional to r^2 .

Which quantity is best represented on the y -axis of the graph?

- A Period of a satellite orbiting the planet.
- B Gravitational potential of the planet.
- C Gravitational field strength of the planet.
- D Kinetic energy of a satellite orbiting the planet.

Your answer

[1]

13 Which of the following is a correct unit for gravitational field strength?

- A J kg^{-1}
- B N kg^{-1}
- C $\text{N m}^2 \text{kg}^{-2}$
- D kg m s^{-1}

Your answer

[1]

- 14 Planets **X** and **Y** each have a single moon.

Planet **X** has twice the mass of planet **Y**. The orbital radius of the moon around planet **X** is three times the orbital radius of the moon around planet **Y**.

The gravitational potential of the planet **X** is V_X at the position of its moon. The gravitational potential of the planet **Y** is V_Y at the position of its moon.

What is the value of the ratio $\frac{V_X}{V_Y}$?

- A 0.22
- B 0.67
- C 1.50
- D 6.00

Your answer

[1]

- 15 A satellite is in a circular orbit around the Earth. The vertical **height** of the satellite above the surface of the Earth is 3200 km. The radius of the Earth is 6400 km.

What is the ratio

$\frac{\text{weight of satellite in orbit}}{\text{weight of satellite on the Earth's surface}}$?

- A 0.25
- B 0.44
- C 0.50
- D 0.67

Your answer

[1]

- 16 A satellite of mass 3000 kg moves from a parking orbit of radius 6800 km to a geostationary orbit of radius 42 000 km. The mass of the Earth is 6.0×10^{24} kg.

What is the magnitude of the change in gravitational potential?

- A. 8.4 J kg^{-1}
- B. $2.5 \times 10^4 \text{ J kg}^{-1}$
- C. $4.9 \times 10^7 \text{ J kg}^{-1}$
- D. $1.5 \times 10^{11} \text{ J kg}^{-1}$

Your answer

[1]

- 17 Scientists are planning to launch a rocket from the surface of the Earth into an orbit at a distance of 18000 km above the centre of the Earth. The radius of the Earth is 6400 km and it has mass 6.0×10^{24} kg.

What is the minimum work done to move the 150 kg mass of the rocket into this orbit?

- A. -13×10^5 J
- B. -6.0×10^5 J
- C. $+6.0 \times 10^5$ J
- D. $+13 \times 10^5$ J

Your answer

[1]

18(a) An asteroid of mass 10 kg is in a circular orbit around the Sun at a distance of 3.5 AU from the centre of the Sun.

$$1 \text{ AU} = 1.5 \times 10^{11} \text{ m}$$

- i. Use Kepler's third law to calculate the period of the orbit of the asteroid in years.

An object orbiting the Sun at a distance of 1 AU has a period of 1 year.

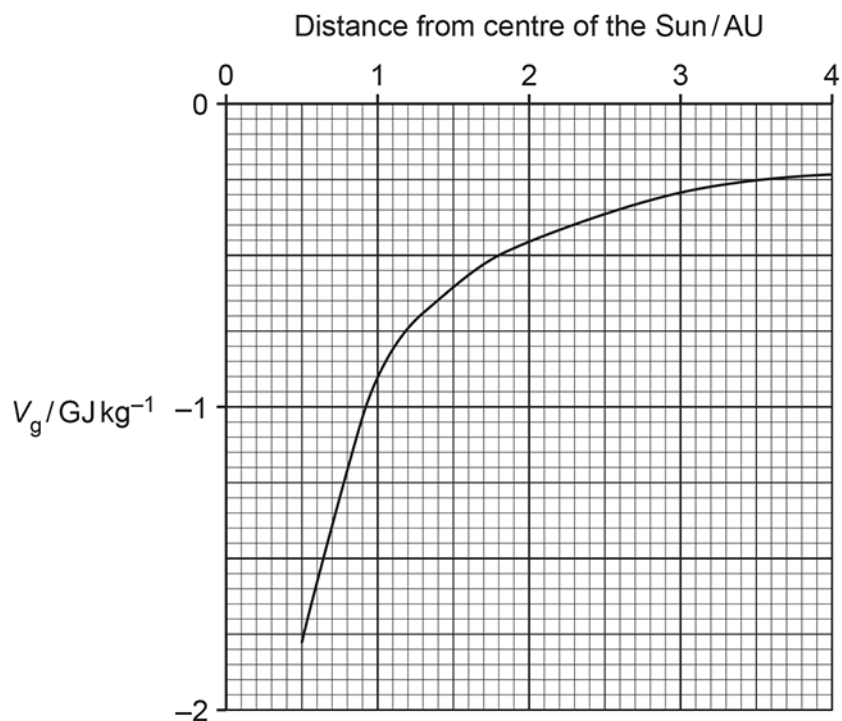
period = years [2]

- ii. Calculate the kinetic energy of the asteroid.

kinetic energy = J [3]

- (b) After a collision the asteroid falls towards the Sun.

The graph below shows the variation of gravitational potential, V_g , due to the Sun with the distance from the centre of the Sun.



- i. Use the graph to calculate the change in gravitational potential energy of the satellite from the moment of collision until it crosses the Earth's orbit.

change in gravitational potential energy = GJ [2]

- ii. Eventually, the asteroid falls into a new orbit around the Sun. This orbit is elliptical.

State the other feature of this orbit predicted by Kepler's first law.

[1]

19(a) An aircraft is flying horizontally at an altitude of 10 000 m and a constant speed of 220 m s^{-1} .

While maintaining this speed, the controls are adjusted so that the aircraft is in **projectile** motion until the altitude is 7500 m.

Throughout the projectile motion the passengers experience apparent weightlessness.

- i. A condition for projectile motion is constant vertical acceleration. State the other condition.

[1]

- ii. Show that t , the time taken to reach the new altitude of 7500 m, is about 23 s.

[2]

- iii. Determine the magnitude and direction of the velocity of the aircraft at time t .

velocity = m s^{-1}
angle to the vertical $^{\circ}$ [4]

- iv. Explain why the engines of the aircraft must provide thrust during the projectile motion.

[2]

(b)

- i. The average radius of the Earth is 6400 km.

Show that the ratio below is greater than 0.99.

$$\frac{\text{gravitational field strength at 10 000 m altitude}}{\text{gravitational field strength at the surface of the Earth}}$$

[2]

- ii. Explain why the passengers are described as 'weightless' during the projectile motion of the aircraft.

[2]

20(a) A satellite in a geostationary orbit around the Earth appears to remain at the same point in the sky when viewed from the ground.

i. State **one** condition required for an orbit to be geostationary.

----- [1]

ii. Calculate the orbital radius of the geostationary satellite. The mass of the Earth is 6.0×10^{24} kg.

orbital radius = m [3]

- (b) A satellite of mass m is in a circular orbit around a planet of mass M . The radius of the orbit from the centre of the planet is r .

The gravitational potential V_g at a point a distance r from the centre of the planet is given by the equation

$$V_g = -\frac{GM}{r}.$$

- i. By considering the cause of the centripetal force on the satellite, show that the kinetic energy of the satellite is equal to half the magnitude of its gravitational potential energy.

[2]

- ii. A tiny satellite of mass 1.0 kg is to be launched from rest from the surface of the Earth into a low Earth orbit. The gravitational potential at any point in this orbit is -56 MJ kg^{-1} .

The value of the gravitational potential at the Earth's surface is -63 MJ kg^{-1} .

Show that the satellite must gain more than 30 MJ of **total** energy to achieve and remain in orbit.

[2]

- (c) Large satellites are often launched by rockets from sites near the equator. The rotation of the Earth increases the initial kinetic energy of the rocket and satellite.

A new strategy is to launch using a smaller rocket from a high flying aircraft.

Using the information in part (ii) of the previous question and the data below, evaluate the advantages and limitations of this strategy. Use calculations to support your evaluation

Rotational speed at the equator	460 m s^{-1}
---------------------------------	------------------------

Typical aircraft operating altitude	10,000 m
Aircraft cruise velocity (relative to the ground)	230 m s ⁻¹

[6]

21 A continuous stream of particles called a solar wind flows from the surface of the star into the surrounding space.

These particles include helium nuclei of mass 6.6×10^{-27} kg.

Assume that the atmosphere is modelled as an ideal gas.

i. Show that the typical kinetic energy of a helium nucleus in the atmosphere is about 10^{-16} J.

[2]

ii. The gravitational potential energy of a helium nucleus in the outer layer of the star is -2.3×10^{-16} J.

Calculate the gravitational potential energy U at the maximum distance from the star that a helium nucleus could reach.

$U = \dots\dots\dots$ J [1]

iii. Calculate the distance from the centre of the star reached by this helium nucleus.

distance = $\dots\dots\dots$ m [3]

iv. Explain why the star has a solar wind that reaches a much greater distance from the star than found in (iii).

----- [1]

22(a)

A planet of mass m is in a circular orbit around a star of mass M .

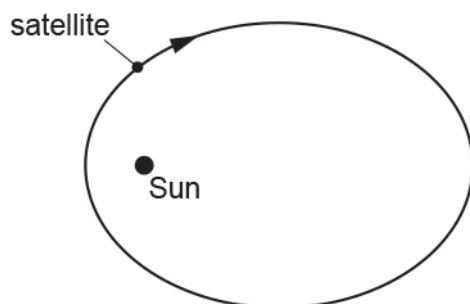
Use the equation for Newton's law of gravitation and your knowledge of circular motion to show that the relationship between the orbital period T of the planet and its orbital radius r is $T^2 \propto r^3$.

[3]

(b) The Solar Orbiter satellite was launched in February 2020.

This satellite moves around the Sun in an elliptical orbit with a period of 168 days.

The diagram below shows the elliptical orbit of this satellite.



The closest distance of the satellite to the Sun is 4.20×10^{10} m and its furthest distance from the Sun is 1.37×10^{11} m.

The mass of the Sun is 2.0×10^{30} kg and the mass of the satellite is 209 kg.

i. The Earth has a mean orbital distance of 1.50×10^{11} m around the Sun and an orbital period of 365 days.

Use **Kepler's third law** to calculate the mean orbital distance of the satellite from the Sun.

distance = m [2]

- ii. The total kinetic and gravitational potential energy of the satellite in its orbit remains constant.

Calculate the change in the kinetic energy of the satellite as it travels from its furthest point from the Sun to its closest point to the Sun.

change in kinetic energy = J [3]

- iii. Suggest why the total energy of the satellite in its orbit around the Sun is not the same as the total energy of the satellite during its launch from the surface of the Earth.

----- [1]

- 23 Algol is a triple-star system, with stars Aa1, Aa2 and Aa3 orbiting each other.
This triple-star is 90 light-years from the Earth.

Here is some data on the star Aa1.

- radius = $(1.90 \pm 0.14) \times 10^9$ m
- mass = $(6.31 \pm 0.42) \times 10^{30}$ kg.

Calculate the gravitational field strength g at the surface of Aa1 to 3 significant figures.

Include the absolute uncertainty in your answer. Assume that the other stars of the system exert negligible gravitational force on Aa1.

$$g = \dots\dots\dots \pm \dots\dots\dots \text{ N kg}^{-1} \text{ [4]}$$

24 The diagram below shows the Earth in space.



- i. On the diagram above, draw a minimum of **four** gravitational field lines to map out the gravitational field pattern around the Earth.

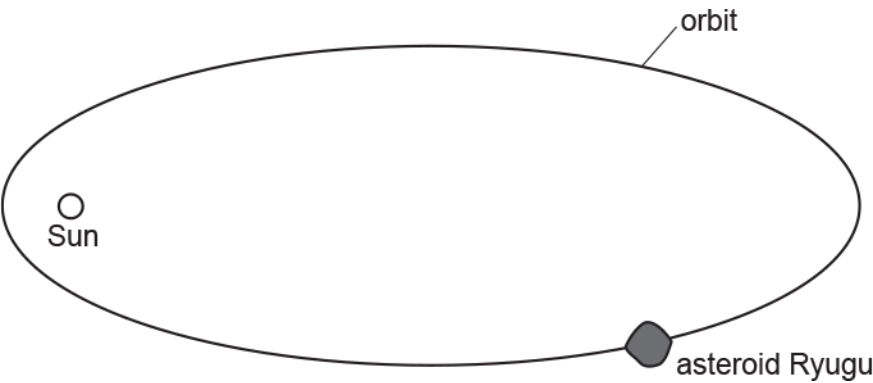
[1]

- ii. On the same diagram above, show **two** different points where the gravitational potential is the same. Label these points **X** and **Y**.

[1]

25(a) In June 2018, the spacecraft Hayabusa2 arrived at an asteroid called Ryugu.

The asteroid orbits the Sun in an elliptical orbit as shown below.



The diagram is **not** drawn to scale.

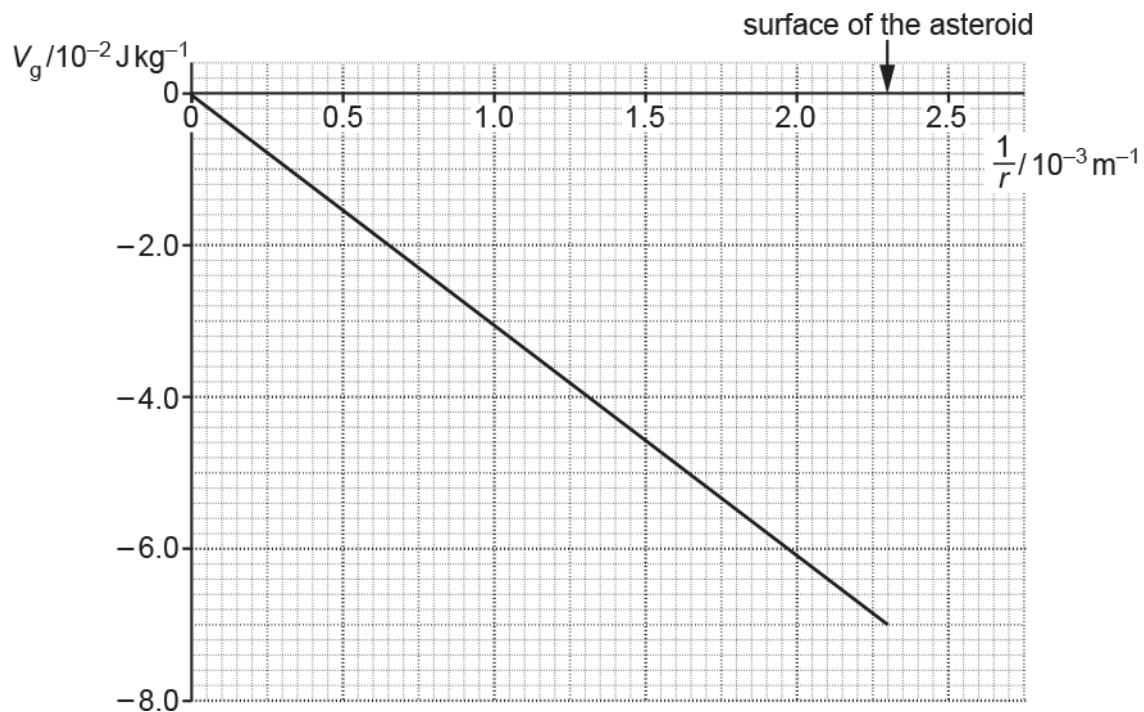
(i) Indicate with a letter **X** on the orbit where the asteroid would be moving at maximum speed.

[1]

1. Use Kepler’s **second law** to explain your answer to (a)(i) .

[2]

(b) The gravitational potential at a distance r from the centre of the asteroid Ryugu is V_g . The graph of V_g against $\frac{1}{r}$ for the asteroid is shown below.



i. Define gravitational potential .

[1]

ii. Show that the magnitude of the gradient of the graph is equal to GM , where M is the mass of the asteroid and G is the gravitational constant.

[1]

iii. Use the gradient of the graph to show that the mass M of the asteroid is about $4.6 \times 10^{11} \text{ kg}$.

$M = \dots\dots\dots \text{ kg}$ [2]

- (c) In October 2018, the probe Mobile Asteroid Surface Scout (MASCOT) was released from **rest** from the Hayabusa2 spacecraft from a distance of 600 m from the centre of the asteroid.

Assume that the spacecraft was stationary relative to the asteroid when MASCOT was dropped.

Use information from (b) to calculate the speed of the impact v when MASCOT landed on the surface of the asteroid.

$$v = \dots\dots\dots \text{ m s}^{-1} \text{ [3]}$$

- 26 Kepler's third law can be applied to a satellite in a geostationary orbit around the Earth.

- i. Complete the equation for Kepler's third law below.
You do not need to define any of the terms.

$$\dots\dots\dots = \frac{4\pi^2}{GM} \dots\dots\dots \text{ [1]}$$

- ii. The mass of Earth is 6.0×10^{24} kg.
Calculate the radius of the circular path of a satellite in a geostationary orbit around the Earth.

$$\text{radius} = \dots\dots\dots \text{ m [2]}$$

27 The galaxies in the Universe may be assumed to be distributed uniformly through space.

In this model, the separation between two neighbouring galaxies is 1.4×10^{23} m and each galaxy occupies a cube of space of volume 2.7×10^{69} m³ as shown in Fig. 24.2.

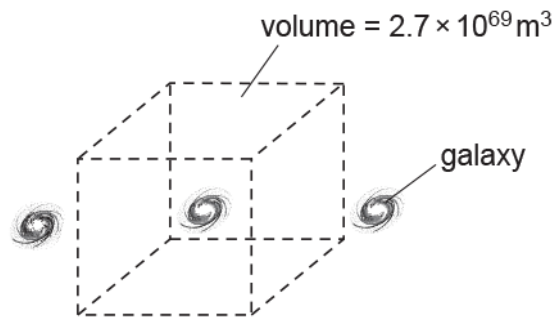


Fig. 24.2

There are on average 10^{11} stars in each galaxy and the mass of an average star is about 2.0×10^{30} kg.

- i. Estimate the gravitational force between two neighbouring galaxies.

force = N [2]

- ii. Show that the mean density of the Universe is about 7×10^{-29} kg m⁻³.

[1]

- iii. Suggest why the actual mean density of the Universe is different from the value calculated in (ii).

[1]

Phobos is one of the two moons orbiting Mars. Fig. 17.1 shows Phobos and Mars.

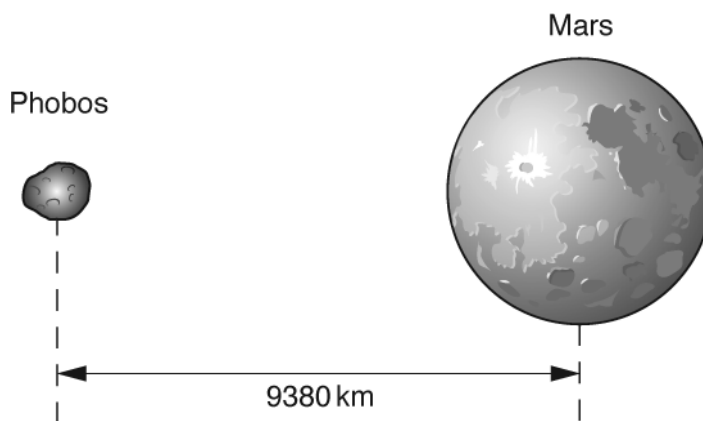


Fig. 17.1

The orbit of Phobos may be assumed to be a circle. The centre of Phobos is at a distance 9380 km from the centre of Mars and it has an orbital speed $2.14 \times 10^3 \text{ m s}^{-1}$.

- i. On Fig. 17.1, draw an arrow to show the direction of the force which keeps Phobos in its orbit. [1]
- ii. Calculate the orbital period T of Phobos.

$T =$ _____ s [2]

- iii. Calculate the mass M of Mars.

$M =$ _____ kg [3]

29(a)

Write an expression for the gravitational potential V_g at the surface of a planet of mass M and radius r .

[1]

(b) The table below shows some data for Mercury and Pluto.

	Mass / kg	Radius / m	Mean distance from Sun / m
Mercury	3.30×10^{23}	2.44×10^6	57.9×10^9
Pluto	0.131×10^{23}	1.19×10^6	5910×10^9

i. Show that the escape velocity v of a gas molecule on the surface of Pluto is given by the equation

$$v = \sqrt{\frac{2GM}{r}}$$

where M is the mass of Pluto and r is its radius.

[2]

i. Calculate the escape velocity v of gas molecules on the surface of Pluto.

$v =$ _____ m s^{-1} [1]

ii. Explain why Mercury has no atmosphere whilst Pluto still has a thin atmosphere. Use data from the table to support your explanation.

[3]

- 30 This question is about a simple pendulum made from a length of string attached to a mass (bob). For oscillations of small amplitude, the acceleration a of the pendulum bob is related to its displacement x by the expression

$$a = -\left(\frac{g}{L}\right)x$$

where g is the acceleration of free fall and L is the length of the pendulum.

The pendulum bob oscillates with simple harmonic motion.

A student conducts an experiment in the laboratory to investigate the small amplitude oscillations of a pendulum of a mechanical clock. Each 'tick' of the clock corresponds to **half** a period.

- i. Show that the length of the pendulum required for a tick of 1.0 s is about 1 m.

[2]

- ii. If the pendulum clock were to be used on the Moon, explain whether this clock would run on time compared with an identical clock on the Earth.

[2]

31(a) The apparatus shown in Fig. 20.1 is used to investigate the variation of the product PV with temperature in the range 20 °C to 100 °C. The pressure exerted by the air is P and the volume of air inside the flask is V .

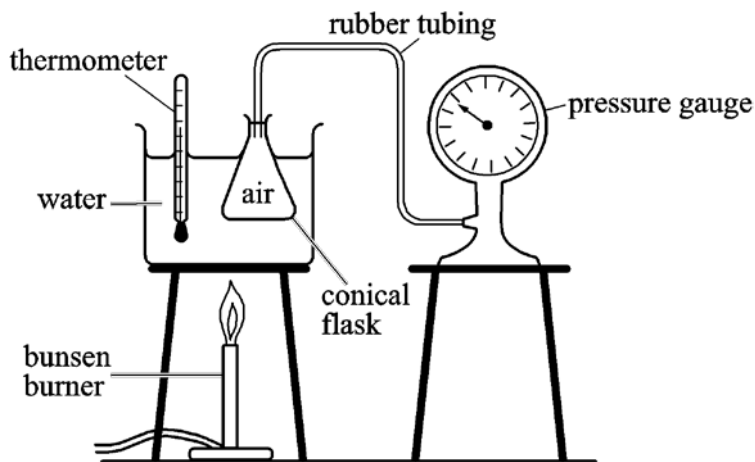


Fig. 20.1

Describe how this apparatus can be set up and used to ensure accurate results.

[4]

- (b) An investigation similar to that shown in Fig. 20.1 gives measurements of the pressure P , volume V and temperature θ in degrees Celsius of a fixed mass of gas.

The results are used to plot the graph of PV against θ shown in Fig. 20.2.

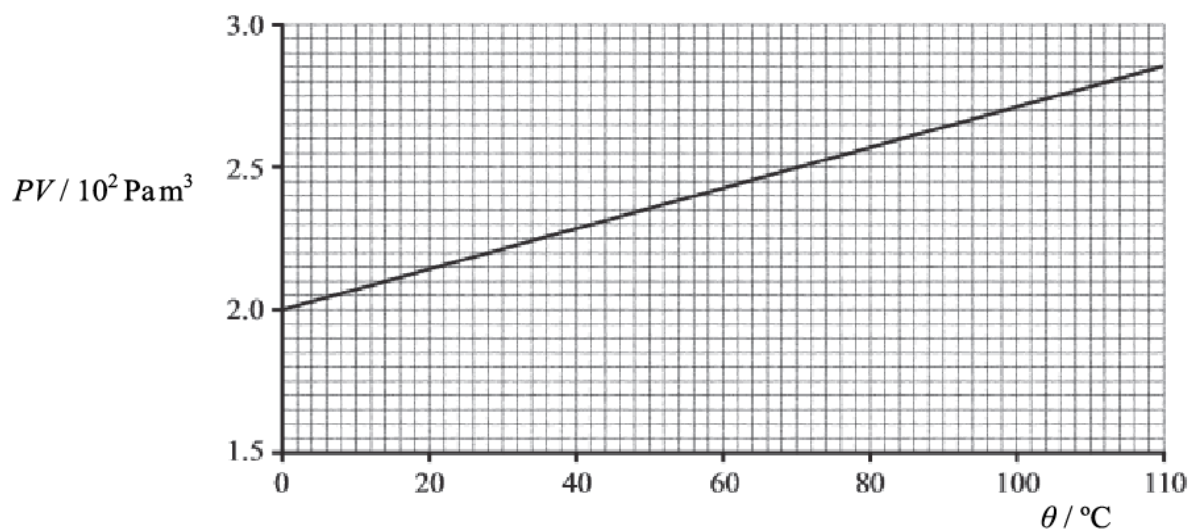


Fig. 20.2

- i. Explain, in terms of the motion of particles, why the graph does **not** go through the origin.

----- [2]

- ii. The mass of a gas particle is 4.7×10^{-26} kg. Use the graph in Fig 20.2 to calculate
1. the mass of the gas

mass = _____ kg

2. the internal energy of the gas at a temperature of 100 °C.

internal energy = _____ J [4]

32 Hubble's law can be used to estimate the age of the universe. Fig. 23 shows some of Hubble's early measurements of nearby galaxies plotted on a v against d graph, where v is the recessional speed of a galaxy and d is its distance from us. Measurements of distant galaxies taken over the last 85 years have refined the value of H_0 to be $68 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

- i. Suggest why measurements for our nearest galaxies can deviate from the current Hubble's law trend line.

----- [1]

- ii. Suggest why measurements for galaxies at the largest distances deviate from the Hubble's law trend line.

----- [1]

33(a) Observations of the planets led to Kepler's three empirical laws:

1. The orbit of a planet is an ellipse with the Sun at one focus.
2. A line joining a planet and the Sun sweeps out equal areas during equal time intervals.
3. The square of the orbital period T is proportional to the cube of the average orbital radius r .

With the help of a labelled diagram, illustrate Kepler's second law for the planets in our Solar System.

[2]

(b) Fig. 22 shows the elliptical orbit of a planet around the Sun.

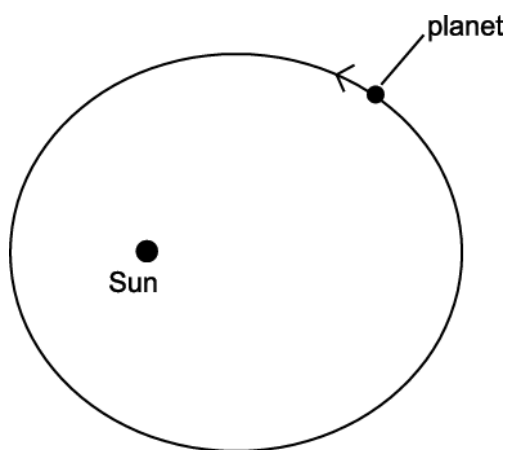


Fig. 22

Draw the gravitational force acting on the planet at the position shown in Fig. 22.

[1]

(c) Three exoplanets orbit the star KIC 11442793. Measurements of average orbital radius r and period T for the exoplanets are shown in the table.

r / AU	T / days
0.0881	8.719
0.520	124.9
0.996	331.6

It is suggested that the relationship between T and r is given by Kepler's third law:

$$T^2 \propto r^3$$

Propose and carry out a test to check if the relationship is true for the three exoplanets.

Test proposed:

Working:

Conclusion:

[3]

34(a) Explain how Newton’s law of gravitation is applied between two non-spherical asteroids.

----- [1]

(b) A satellite moves in a circular orbit of radius 15 300 km from the centre of the Earth.

i. State one of the main benefits satellites have on our lives.

----- [1]

ii. Calculate the gravitational field strength g at the radius of 15 300 km.

$g =$ _____ N kg⁻¹ [2]

iii. Calculate the period of the orbiting satellite.

period = _____ s [3]

(c) Determine the average density of the Earth. The radius of the Earth is 6400 km.

average density = _____ kg m⁻³ [3]

35 A satellite is in a circular geostationary orbit around the centre of the Earth. The satellite has both kinetic energy and gravitational potential energy.

The mass of the satellite is 2500 kg and the radius of its circular orbit is 4.22×10^7 m.
The mass of the Earth is 5.97×10^{24} kg.

- Describe some of the features of a geostationary orbit.
- Calculate the **total** energy of the satellite in its geostationary orbit.

[6]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1			A	1	Examiner's Comments The key here is to use the formula for gravitational force ($F = \frac{GMm}{r^2}$) to find the distance between the centres of the spheres. This gives 100.2 m. The question asks for the distance between the surfaces of the two spheres, which is 20 m less than the distance between their centres. Hence A is the correct answer.
			Total	1	
2			C	1	Examiner's Comments By considering the GPE at the planet's surface of $-GMm / r$ and the KE of $\frac{1}{2} m v^2$, the condition for 'escape' is that the total energy is zero at infinity. This gives the condition $GMm / r = \frac{1}{2} m v^2$. The gravitational field strength at the surface of the planet is GMm / r^2 i.e. g, giving $gr = \frac{1}{2} v^2$. When rearranged this gives answer C.
			Total	1	
3			B	1	
			Total	1	
4			D	1	Examiner's Comments The question is asking for the candidate to calculate the speed of an object in a circular orbit with radius 42 400 km, i.e. 42.4×10^6 m. To do this, the candidate equates the gravitational force, $FG = GMm/r^2$ with the centripetal force required for that orbit, $FC = mv^2/r$, where M is the mass of the Earth and m is the mass of the satellite. Rearranging gives $v = \sqrt{GM/r}$. Substituting the correct values for G, M and r will give answer D.
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
5			C	1	<u>Examiner's Comments</u> Most candidates suggested that angular frequency and angular velocity do not have the same or equivalent units. The correct answer is C, because gravitational potential is measured in J kg^{-1} and kinetic energy is measured in J.
			Total	1	
6			C	1	
			Total	1	
7			B	1	
			Total	1	
8			B	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
9			B	1	<u>Examiner's Comments</u> Answering this question needs knowledge of Kepler's Third Law and the formulae for gravitational potential and gravitational field strength, all of which are given in the data, formulae and relationships booklet. This relationship cannot be gravitational potential or gravitational field strength, as quantity y increases with distance from the Sun. By equating the gravitational force on a planet with the centripetal force, it can be shown also that $(\text{orbital speed})^2$ and orbital radius are inversely proportional. This graph does not show an inversely proportional relationship. The formula sheet says that the square of a planet's period is directly proportional to the cube of the planet's orbital radius. In other words, the relationship shows that as orbital radius increases, so does the period, but not directly proportionally. This is the relationship shown on the graph, giving answer B.
			Total	1	
10			D	1	<u>Examiner's Comments</u> Near to the surface of the Earth, the gravitational field is approximately uniform. This means that they are parallel and equally spaced. Gravitational field lines in general show the direction of the force on a small mass. This makes all 3 statements are correct, giving the answer D.
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
11			C	1	Examiner's Comments This question requires the candidate to calculate the gravitational potential energy when $r = 6.4 \times 10^3$ m and again when $r = 7.6 \times 10^3$ m. The difference of those two energies gives answer C. About two thirds of all candidates got this correct.
			Total	1	
12			C	1	Examiner's Comments This question proved particularly straightforward and accessible to nearly all candidates.
			Total	1	
13			B	1	Examiner's Comments Virtually all questions showed a positive discrimination, except for question 10. The questions themselves require careful inspection, as crucial information that could lead to the exclusion of many options can be obtained reducing the need for calculation and guessing. Underlining or circling key points may help candidates to converge towards the correct responses. Candidates should ensure that all letters are clearly formed. If there is a need to amend an answer crossing through the incorrect answer and writing the correct answer adjacent to the box will help avoid any potential for misunderstanding by the examiner. This question proved particularly straightforward and accessible to nearly all candidates.
			Total	1	
14			B	1	
			Total	1	
15			B	1	
			Total	1	

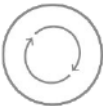
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
16			C	1	
			Total	1	
17			C	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
18	a	i	$T_{\text{asteroid}}^2 = 3.5^3$ $T = 6.5 \text{ (yr)}$	C1 A1	Accept substitution of given formula with AU in metres (giving time in seconds) Answer to 3sf is 6.55 (yr) Examiner's Comments While some candidates missed off the correct power for either time or orbital radius, many knew which relationship to use. The challenge here was to get the right units for both time and orbital radius. By using the units of AU and years, it was possible simply to write that $T^2 = R^3$ and to work from there. The full relationship involving the mass of the Sun and working in SI units was less straightforward; however, many candidates managed to complete this correctly. Errors were made with the mass (using 10 kg instead of the mass of the Sun) or the radius (which needed to be in metres) for the time to be calculated in seconds (requiring a further conversion to be able to give the period in the required unit of years.)
		ii	$R = 3.5 \text{ AU} = 3.5 \times 1.5 \times 10^{11} = 5.3 \times 10^{11} \text{ (m)}$ $v = 2\pi R/T = 16 \text{ (kms}^{-1}\text{)}$ $\text{KE} = \frac{1}{2}mv^2 = 1.3 \times 10^9 \text{ J}$	C1 C1 A1	ECF from a(i) for T Note: $v = R/T$ is XP Allow ECF from v Allow $1.2 \times 10^9 \text{ (J)}$ if no premature rounding in calculation Answer is $1.26 \times 10^9 \text{ (J)}$ to 3sf Alternative method $mv^2/r = GMm/r^2 \text{ (C1)}$ $\frac{1}{2}mv^2 = m \times \frac{1}{2}GM/r = \frac{1}{2} \text{ of GPE (C1)}$ $= 1.3 \times 10^9 \text{ J from graph (A1)}$ Examiner's Comments The 'error carried forward' rule was applied here in the case of the wrong value for the orbital period and again if the orbital velocity was incorrectly calculated. Otherwise this question was successfully completed.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	b	i	Potential difference from graph = $0.9 - 0.25 = 0.65 \text{ GJ kg}^{-1}$ GPE = mass $\times$ potential difference = 6.5 GJ	C1 A1	If calculated value is 0.89-0.25 ignore sign Allow ECF from the graph <u>Examiner's Comments</u> Candidates found this challenging. The graph gives the gravitational potential. The correct approach is to find the difference in the gravitational potential at 1 AU and at 3.5 AU and then multiply this difference by the mass of the satellite.  Assessment for learning Candidates should learn the difference between gravitational potential, potential energy, force and field strength. They are all similar ideas with only minor differences in their formulae. A similar approach could be taken with the electric field equivalent elsewhere in the specification.
		ii	The Sun is at one focus	B1	<u>Examiner's Comments</u> A large proportion of candidates successfully identified that the other feature was that the Sun lay at one of the foci of an orbit.
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
19	a	i	constant horizontal speed/constant horizontal (component of) velocity/zero horizontal force (however expressed)/zero horizontal acceleration	A1	Allow any alternate wording Examiner's Comments There were many ways to express the idea that the horizontal velocity should stay constant.
		ii	Use of $s = \frac{1}{2} at^2$ with $s=2500$ and with $a=9.8(1)$ or $a=g$ consistent algebra that arrives at $t^2 = 2 \times 2500 / g$ and evaluation of t as a decimal e.g. $t = 22.6(\dots)$	M1 A1	Allow alternate using $v^2 = u^2 + 2as$ and then $v = u + at$ NOT $s = \frac{1}{2} vt$ with $v=220$ found by $v = u + at$ (i.e. use of $t = 23$ to find v) Allow $t^2 = 509.6\dots$ for $t^2 = 2 \times 2500/g$ Allow $t = \sqrt{(2 \times 2500/g)}$ for $t^2 = 2 \times 2500/g$ Allow $9.8(1)$ for g Responses with $u=220$ are XP Examiner's Comments This question tests the idea of independence of horizontal and vertical motion. There are lots of letters involved in the algebra that are applicable to both motions, e.g. u, v, a, and s. Successful candidates made the distinction between horizontal and vertical motion as clear as possible, either by using a two-column table with the headings 'vertical' and 'horizontal' or by using V and H subscripts on the relevant letters. Once candidates got to the starting point of a displacement of 2500 m with zero initial vertical velocity and an acceleration of 'g', they often arrived at the correct answer.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	Vertical $v = gt = 220 \text{ ms}^{-1}$ $v = (220^2 + 220^2)^{0.5}$ $v = 310 \text{ (ms}^{-1}\text{)}$ Angle = 45°	C1 C1 A1 A1	ECF for their value of t ECF for their value of vertical v Allow answer that rounds to 3sf to be in range 310-315 (m s⁻¹) Allow “220” for “maintaining this speed” NOT left as a surd NOT left as a trigonometric expression Allow answer in range 44-45 Allow answer in range 135-136 NOT left as a trigonometric expression NOT left as a surd ECF for their vertical velocity and use of trigonometry NOTE acute answer above 45.0 is incorrect <u>Examiner's Comments</u> A diagram of the two components of the velocity here was not mandatory; however, it makes it clear in the mind of the candidate what they are doing. The most successful candidates drew a triangle with the two components of velocity with an angle indicated between the vertical and the resultant velocity. They were then clear about how to work out the resultant velocity (Pythagoras) and the required angle. Both acute and obtuse angles to the vertical were acceptable. Acute angles over 45 degrees indicated an error in the trigonometry and so the last mark was unavailable. Few candidates were able to correctly interpret the question's description of ‘maintaining this speed’ to suggest that the velocity of the plane was 220 m s^{-1} throughout.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iv	Reference to drag or air resistance or friction Thrust required for zero horizontal net force / thrust = drag	B1 B1	Accept reference to work done against resistive forces (either named or unnamed) Accept thrust allow to keep constant: vertical acceleration / horizontal velocity / horizontal speed Accept AW for “thrust” e.g. driving force <u>Examiner’s Comments</u> There were lots of ways the candidates used to express the right ideas. As in previous series, phrases such as ‘balanced forces’ were not specific enough. ‘Thrust is required for horizontal net force’ is one example of using more specific language to gain credit here.
	b	i	g is proportional to $1/r^2$ $(6400 / 6410)^2 > 0.991$	M1 A1	Allow $g=GM/r^2$ Accept calculation of value of g at altitude Accept calculation of g at altitude divided by g at ground to give a number greater than 0.991. NOTE: 6400/6410 giving 0.9984 is XP NOTE: ratios that use the same value of r is XP <u>Examiner’s Comments</u> The mark scheme indicates two alternative approaches to this item, which covered most attempts by candidates. Common errors, however, included missing the ‘squared’ sign off the distance from the centre of the Earth either on paper or on the calculator, or getting the fraction upside-down.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	Aircraft and passengers have the same acceleration(g) No contact force(s on passenger from seat/body of the aircraft)	B1 B1	Accept passenger is (also) moving as a projectile Accept relative acceleration between aircraft and passenger is zero Accept reaction for contact NOT: idea that there no forces on passenger <u>Examiner's Comments</u> There were two separate ideas here. Firstly that the objects are travelling at the same acceleration (i.e. 'g') and the effect this has on the normal contact force between the plane and the passenger (i.e. there is none).  Misconception Candidates often think that because they've seen things floating on video coverage of astronauts in space that there is no resultant force acting on them. This is a misconception. There is no <i>relative</i> motion but both the ISS and the astronauts inside them are travelling very quickly in Low Earth Orbit. They can only do this because of the centripetal force provided by gravitation with the Earth.
			Total	13	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
20	a	i	Period equal to one day/24 hours/rotational period of Earth Or Rotation in same direction as the rotation of the Earth (on its axis) Or Equatorial orbit Or radius of orbit is approx 42 000 km Or height of orbit above Earth's surface is approx 36 000 km Or zero velocity relative to Earth's surface	B1	NOT 'is in the same place in the sky' Examiner's Comments In previous series, candidates needed to explain at least three conditions for a geostationary orbit for complete credit. This item, though, is considerably simpler: most candidates scored a mark here. There was some confusion about terminology with those candidates that did not score the mark. Often this was muddling up the time of orbit, the time of rotation of the Earth and the time of an orbit of the Earth around the Sun.
		ii	Use of $T=3600 \times 24$ (or 86,400 from data book) Substitution into $T^2 = \frac{4\pi^2}{GM} r^3$ i.e. $(86400)^2 = (4\pi^2 / (G \times 6.0 \times 10^{24})) r^3$ $r = 4.2 \times 10^7$ (m)	C1 C1 A1	Examiner's Comments Most candidates correctly used the equation representing Kepler's Third Law to find the orbital radius. Common errors included mis-converting or mis-remembering the time for one complete orbit.
	b	i	$\frac{GMm}{r^2} = \frac{mv^2}{r}$ $\frac{1}{2}mv^2 = \frac{1}{2} \frac{GMm}{r}$	M1 A1	Allow omission of 'm' on both sides of equation (gravitational field strength = centripetal acceleration) Cancelling/rearrangement/identification of GMm/r as GPE Examiner's Comments Many candidates made a good start by equating the formula for gravitational force with the expression for centripetal motion. Others that assumed that g of 9.81 m s^{-2} did not score any marks. The simplest way to arrive at the correct expression was to identify GM/r in the gravitational force formula, to rearrange and then multiply both sides by $\frac{1}{2}$. Approximately $\frac{1}{2}$ of all candidates that responded got as far as this gaining both marks.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	(Increase in) GPE = $(-56 - -63 \text{ MJ}) = 7 \text{ (MJ)}$ or (Increase in) KE = $0.5 \times 56 = 28 \text{ (MJ)}$ Sensible reasoning, e.g. $7+28 > 30$	M1 A1	Allow evaluation of total energy of 35 (MJ) Examiner's Comments Many candidates correctly determined how much GPE the satellite needed to gain i.e. 7 MJ in order to reach orbit from -63 MJ to -56 MJ. To find the KE when in orbit, candidates needed to use the result from the previous part of the question. This explains why the in orbit, the KE required is $\frac{1}{2}$ of 56 MJ i.e. 28 MJ. A small fraction of candidates successfully accomplished this step. This means the total energy gain is the sum of 28 MJ and 7 MJ i.e. 35 MJ.
	c		Level 3 (5–6 marks) Correct calculations, and advantages and limitations discussed. <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Correct calculations and an advantages discussed or Correct calculations and a limitation discussed. <i>There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Attempted calculations or a single correct calculation or incomplete explanations of advantages and/or limitations. <i>There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.</i> 0 mark <i>No response or no response worthy of credit.</i>	B1 × 6	Use level of response annotations in RM Assessor Ignore general knowledge answers e.g. accidents, cost, politics Allow references to energy, energy per unit mass or potential as interchangeable Allow ecf from candidate's value for total energy per unit mass in orbit from (i.e. 35 MJ) 22bii or use of 30 (MJ) Indicative scientific points may include: Calculations • (Minimum) additional KE from aircraft=26 kJ • Additional GPE from aircraft = 100 kJ • Additional KE from equatorial launch=110 kJ • GPE calculated by mgh as an acceptable approximation • (Without taking Earth's rotation into account,)KE at equator is about 4x aircraft KE • GPE calculated from $vg = (-)GM/r$ • Total energy = $0.5GM/r$ Advantages • Aircraft launch provides KE and GPE • Aircraft velocity will be higher than 230 depending on where the aircraft takes off. • Less (rocket) fuel required

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			• Aircraft launch has similar/slightly larger energy to equatorial • Equatorial launches can only happen in limited locations/aircraft launches can take place almost anywhere Limitations • Aircraft launches only suitable for small satellites. • Effects (for either) only significant for near earth orbits/low altitudes • Either launch provides very small fraction (less than 1%) of the energy required Examiner's Comments Many candidates found this difficult to access. A solely qualitative evaluation was limited to level 1 (2 marks). Exemplar 2 The initial kinetic energy + GPE will give the total energy gained by the satellite. On earth $\rightarrow \frac{1}{2} \times 1 \times 460^2 = 0.1058 \text{ MJ}$ of kinetic energy On the aircraft, $\frac{1}{2} \times 1 \times 230^2 = 0.02645 \text{ MJ}$ However, on the aircraft, the GPE will be higher, and the satellite will need to gain less GPE. $\frac{6.6 \times 10^{-4} \times 6 \times 10^{24} \times 1}{r_{\text{earth}} + 10000} = \text{GPE advantage.}$ In conclusion, although the aircraft would slightly reduce energy required for GPE increase, it would reduce initial kinetic energy by a factor of 4, and there would also be energy required to power the aircraft. In Exemplar 2, the candidate has completed a small number of calculations comparing the KE the two methods would raise. There is also a statement that these differ by a factor of 4 along with an attempt at calculating the GPE advantage by launching from the aircraft rather than from the ground. Crucially, there is very little mention of limitation and no supporting calculations. Holistically speaking, therefore, there was not enough evidence to award this candidate Level 3, yet sufficient for a Level 2. Exemplar 3

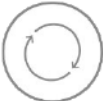
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance												
					<table><tr><td>Rotational speed at the equator</td><td>460 ms⁻¹</td><td>Typical aircraft operating altitude</td><td>10,000 m</td></tr><tr><td colspan="2">Strategy (A)</td><td>Aircraft cruise velocity (relative to the ground)</td><td>230 ms⁻¹</td></tr><tr><td colspan="2"></td><td>Strategy (B)</td><td></td></tr></table> We need 35 MJ of energy per kg. In Strategy A, we start with 460 ms⁻¹, resulting in $\frac{1}{2}(460)^2 = 105800 \text{ J} = 0.11 \text{ MJ}$ of energy per kg to create energy. In Strategy B, we have 567000 J (0.567 MJ) on approach in GPE of mgh = 9.81 (10000) = 98100 J = 0.098 MJ per kg adding the GPE's gravitational yield as constant. We also have a KE of velocity of 230 ms⁻¹ so we cruise over the equator ^{in direction of rotation} we get a speed of (460+230) ms⁻¹ KE = $\frac{1}{2}(460+230)^2 = 230000 \text{ J} = 0.23 \text{ MJ}$ giving us a total energy of about 0.25 MJ per kg, much better than Strategy (A). However the energy is still 6.5% less than the 30 MJ kg⁻¹ of energy needed to get the satellite in orbit. However the energy gain may be better as the Earth's atmosphere is thinner at 10,000 m, so the satellite will be less excited. But still it may not be worth it to get the satellite at 10,000 m. Additional space if required Firstly the aircraft required will be expensive and specialist and require smooth get to 10,000 m, however it does mean that we don't have to build a rocket here launch straight at the equator (more unstable condition) as we can just fly over the equator and take off from. Overall less the problems of getting gain by launching ^{launching} the satellite at the equator are just not significant enough to justify the expensive equipment and design costs; and the extra fuel it's much easier to just add more fuel to the satellite and drop the fuel containers in stages in a ground launch. Only about 0.83% of total energy required. In Exemplar 3, this candidate makes an excellent attempt at every portion of the question. They have taken great effort to make themselves clear with technical language. They have also performed numerous calculations to compare the GPE and KE in the two different scenarios. A further figure appears right at the bottom of the response. The candidate has correctly calculated the percentage of the overall energy required that is supplied by either method. In conclusion, the candidate has used their many correct calculations to weigh up evidence effectively in terms of the advantages and limitations.	Rotational speed at the equator	460 ms ⁻¹	Typical aircraft operating altitude	10,000 m	Strategy (A)		Aircraft cruise velocity (relative to the ground)	230 ms ⁻¹			Strategy (B)	
Rotational speed at the equator	460 ms ⁻¹	Typical aircraft operating altitude	10,000 m														
Strategy (A)		Aircraft cruise velocity (relative to the ground)	230 ms ⁻¹														
		Strategy (B)															
			Total	14													

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
21		i	Use of $KE_{\text{mean}} = \frac{3}{2} kT = \frac{3}{2} \times k \times 4.0 \times 10^6$ $= \frac{3}{2} \times 1.38 \times 10^{-23} \times 4 \times 10^6$ $= 8.3 \times 10^{-17} \text{ J.}$	C1 A1	Allow k Unrounded answer is $8.28... \times 10^{-17}$ Examiner's Comments Candidates answered Question 23 (b) (i) well, provided that they used the correct temperature i.e. that of the stellar atmosphere and not the surface temperature.
		ii	$= -2.3 \times 10^{-16} + 8.3 \times 10^{-17}$ $= -1.5 \times 10^{-16} \text{ J.}$	B1	Allow use of 10^{-16} J from stem to give $-1.3 \times 10^{-16} \text{ J}$ Reject response without negative sign Examiner's Comments The helium nucleus in parts (b) (ii) and (b) (iii) has a KE of 10^{-16} J and a GPE of $-2.3 \times 10^{-16} \text{ J}$. This helium nucleus cannot therefore escape and so the GPE at the further possible point of the helium nucleus must be negative. The GPE at this furthest possible point is where the KE is zero and there has been a gain of 10^{-16} J of GPE, giving the GPE $= -1.3 \times 10^{-16} \text{ J}$, using the values in the question. Of course, we used the exact value calculated in part (b) (i) instead of 10^{-16} J, giving $U = -1.5 \times 10^{-16} \text{ J}$ as the 'correct' answer.
		iii	selection of GMm/r Substitution of values for M,m and energy and rearrangement $3.2 \times 10^{10} \text{ m}$ e.g. $(-) \frac{GMm}{r} = (-) 1.5 \times 10^{-16}$ $r = \frac{6.67 \times 10^{-11} \times 1.1 \times 10^{31} \times 6.6 \times 10^{-27}}{1.5 \times 10^{-16}}$ $= 3.2 \times 10^{10} \text{ m}$	C1 C1 A1	Allow ECF from (b)(ii) NB Use of values in stem gives 3.7×10^{10} Examiner's Comments The helium nucleus in parts (b) (ii) and (b) (iii) has a KE of 10^{-16} J and a GPE of $-2.3 \times 10^{-16} \text{ J}$. This helium nucleus cannot therefore escape and so the GPE at the further possible point of the helium nucleus must be negative. The GPE at this furthest possible point is where the KE is zero and there has been a gain of 10^{-16} J of GPE, giving the GPE $= -1.3 \times 10^{-16} \text{ J}$, using the values in the question. Of course, we used the exact value calculated in part (b) (i) instead of 10^{-16} J, giving $U = -1.5 \times 10^{-16} \text{ J}$ as the 'correct' answer.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iv	Reference to Boltzmann distribution / AW OR Some particles will have greater kinetic energies /lower masses	B1	<u>Examiner's Comments</u> Question 23 (b) (iv) has several acceptable responses yet the predominant correct candidate answers were that the solar wind particles might be lighter (i.e. hydrogen) or that the Maxwell-Boltzmann distribution of KE values in a gas of particles would mean that some particles would have greater than average KE.  Assessment for learning Usually in physics calculations, we use the calculator value all the way through the calculation. Any rounding should take place at the very end of the calculation.
			Total	7	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
22	a		$G \frac{Mm}{r^2} = \frac{mv^2}{r}$ or $G \frac{Mm}{r^2} = mr\omega^2$	M1	Allow any subject
			$v = \frac{2\pi r}{T}$ or $\omega = \frac{2\pi}{T}$	M1	<u>Examiner's Comments</u>
			Substitution and manipulation to give $T^2 = \frac{4\pi^2}{GM} r^3$ (with $\frac{4\pi^2}{GM}$ is constant)	A1	The demonstration of Kepler's 3 rd Law was well-remembered by the vast majority of candidates.
	b	i	$\left(\frac{168}{365}\right)^2 = \left(\frac{r}{1.50 \times 10^{11}}\right)^3$	C1	Ignore calculation of arithmetic mean of data in question
			distance = 8.9×10^{10} (m)	A1	Allow substitution into $T^2 = \frac{4\pi^2}{GM} r^3$ Ignore units for subs into Kepler's law NOT 8.95 or $9(.0) \times 10^{10}$ (m) (mean calculated) <u>Examiner's Comments</u> Fortunately, most candidates saw the reference to Kepler's third law in bold and used it successfully. Only a small fraction found the mean orbital distance by finding the mean of the maximum and minimum distance. While this is mathematically sound, it does not use Kepler's third law and so was not deemed an acceptable answer to the question.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	$(\Delta \text{GPE} = \Delta \text{KE})$ $GMm \left(\frac{1}{4.20 \times 10^{10}} \right) \text{ or } GMm \left(\frac{1}{1.37 \times 10^{11}} \right)$ (change in KE =) $6.67 \times 10^{-11} \times 2.0 \times 10^{30} \times$ $m \left(\frac{1}{4.20 \times 10^{10}} - \frac{1}{1.37 \times 10^{11}} \right)$ change in kinetic energy = $4.6 \times 10^{11} \text{ (J)}$	C1 C1 A1	Allow this mark without the m Allow this mark without the m Allow 2 marks for 2.2×10^9 ; ΔV calculated Ignore sign <u>Examiner's Comments</u> Previous examiner's reports have stated the importance of understanding how gravitational potential energy differences are calculated using the appropriate formula. More candidates got this idea right in this series, which was excellent. $\text{energy} = -\frac{GMm}{r}, \Delta E_k = \left(-\frac{G(2.0 \times 10^{30})(209)}{1.37 \times 10^{11}} \right) - \left(-\frac{G(2.0 \times 10^{30})(209)}{4.20 \times 10^{10}} \right)$ $\Delta E_k = 4.60 \times 10^{11} \text{ J}$ change in kinetic energy = 4.60×10^{11} J [3]
		iii	Description of reasonable effect of Earth has been ignored / work done by fuel (during lift off) / idea that atmosphere has been ignored previously	B1	<u>Examiner's Comments</u> This question provides one of the 'stretch and challenge' marks in this paper. The idea is that the satellite needs first to climb out of the Earth's gravitational well before it can be a satellite of the Sun.
			Total	9	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
23			$g = \frac{G \times 6.31 \times 10^{30}}{(1.90 \times 10^9)^2}$ $g = 117 \text{ (N kg}^{-1}\text{)}$ $2 \times \frac{0.14}{1.90} + \frac{0.42}{6.31} \text{ or } 0.21 \text{ or } 21\%$ (absolute uncertainty =) 25 (N kg⁻¹)	C1 C1 C1 A1	Reject $0.42/(0.14^2) = 21.4$ Note: final answer of $g = 117 \pm 25 \text{ (N kg}^{-1}\text{)}$ with no working scores all 4 marks. Allow correct identification of %(r) and %(m) for 1 mark max if no other marks scored. Allow alternate method using max/min values of m and r that give the correct absolute uncertainty (28)
			Total	4	
24		i	Straight symmetrical radial field lines and correct direction of field	B1	Ignore field lines inside the Earth
		ii	X and Y labelled which should be an equal distance away from the centre of the Earth	B1	Note Judge by eye Allow X and Y both on the surface of the Earth
			Total	2	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
25	a	i	X at closest point on orbit to the Sun	B1	Allow X on the orbit to the <u>left</u> of the Sun
		ii	(When the asteroid orbits the sun a) line segment joining the asteroid to the Sun sweeps out equal areas in equal time (intervals) Longer distance (in orbit for the same time)	B1 B1	Allow this mark on diagram (no labelling required) Allow 'equal area swept in same time'
	b	i	Work done per unit mass to move an object from infinity (to that point)	B1	Not 'work done on 1 kg'
		ii	Manipulation of $V_{(g)} = (-) GM/r$	B1	
		iii	gradient = $(-)30.4$ or equivalent working candidate's gradient or expression = $6.67 \times 10^{-11} \times M$ and M calculated correctly from that gradient $M = 4.6 \times 10^{11}$ (kg)	C1 C1 A0	Allow ± 2 Possible ECF from incorrect gradient Allow any subject
	c		Method 1: Evidence of 2.3×10^{-3} <u>and</u> 600^{-1} or $(2.3 \times 10^{-3})^{-1}$ and 600 $\frac{1}{2} v^2 = 6.67 \times 10^{-11} \times 4.6 \times 10^{11} \times (2.3 \times 10^{-3} - 600^{-1})$ $v = 0.20$ (m s ⁻¹) Method 2: Evidence of 7.0×10^{-2} <u>and</u> 5.1×10^{-2} from graph $\frac{1}{2} v^2 (= \Delta V_{(g)}) = 7.0 \times 10^{-2} - 5.1 \times 10^{-2}$ $v = 0.19$ (m s ⁻¹)	C1 C1 A1 (C1) (C1) (A1)	Possible ECF from (b)(iii) for either value of GM or M Allow $\frac{1}{2} v^2 = 30 \times (2.3 \times 10^{-3} - 600^{-1})$ Note answer can be 0.19 or 0.20 or 0.2 m s ⁻¹ Note answer can be 0.19 or 0.20 or 0.2 m s ⁻¹ Allow correct use of one piece of data arriving at a value for v for 1 mark max
			Total	10	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
26		i	$T^2 = \frac{4\pi^2}{GM} r^3$	B1	
		ii	$86400^2 = (4\pi^2/6.0 \times 10^{24} \times 6.67 \times 10^{-11}) r^3$ (Any subject) radius = 4.23×10^7 (m)	C1 A1	<u>Examiner's Comments</u> The radius of the orbit of a geostationary satellite was found with ease by the majority of candidates by using the formula in the data book or by looking at their response for part 23(b)(i). There were a few ways of generating an arithmetical error, such as using the square root instead of the cube root for the final step, getting the wrong power for the time or by using a time equal to one year instead of one day.
			Total	3	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
27		i	$F = GMm/r^2$ $F = G \times (2.0 \times 10^{41})^2 / (1.4 \times 10^{23})^2$ force = 1.4×10^{26} (N)	C1 A1	Note the mark is for substitution, value of G is not required Ignore: minus sign Allow 1 mark for 1.4×10^4 N ;use of mass of star instead of mass of galaxy. <u>Examiner's Comments</u> While some lower level responses included an attempt to find the gravitational field strength rather than the force most selected the correct formula. After selecting the correct relationship, most candidates could then correctly find the force, provided that they remembered to multiply the masses and square the distance of separation.
		ii	density = $10^{11} \times 2.0 \times 10^{30} / 2.7 \times 10^{69}$ density = 7.4×10^{-29} (kg m ⁻³)	M1 A0	
		iii	Any reasonable answers questioning model such as observed average distance may be different, average mass may be wrong etc.	B1	e.g. black holes, dark energy/matter, expanding universe <u>Examiner's Comments</u> This was a question about challenging the model of the universe. The model takes into account an average mass and average distance of separation, so answers that referred to a variation in masses or distances between galaxies did not score. Higher level responses included that the universe was expanding, so that the distances involved were always changing, or that dark matter was not included in the calculations. There was no indication that candidates were constrained by time in this paper.
			Total	4	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
28		i	Horizontal arrow pointing to the right.	B1	Judgement by eye <u>Examiner's Comments</u> The examiners were quite lenient in this series in terms of the precise direction of the arrow, which should point towards the centre of Mars.
		ii	$2.14 \times 10^3 = \frac{2\pi \times 9380 \times 10^3}{T}$ $T = 2.75 \times 10^4 \text{ (s)}$	C1 A1	Allow 2SF answer Note: $2.75 \dots \times 10^n$ scores 1 mark. <u>Examiner's Comments</u> Around four fifths of candidates got this right. Those that did not either poorly converted the radius from km or used the area rather than the circumference of the orbit.
		iii	$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad \text{or} \quad v^2 = \frac{GM}{r}$ $(2.14 \times 10^3)^2 = 6.67 \times 10^{-11} \times M/9380 \times 10^3$ $M = 6.44 \times 10^{23} \text{ (kg)}$	C1 C1 A1	Allow ecf of answer for T from (a)(ii) Allow 2 SF answer Note: Use of 2.8×10^4 seconds gives 6.3×10^{23} (kg) for 3 marks. Alternative Method for C1C1 $M = 4\pi^2 R^3 / (T^2 G)$ (Databook formula re-arranged with M as subject) $M = 4\pi^2 (9380 \times 10^3)^3 / ((2.75 \times 10^4)^2 \times 6.67 \times 10^{-11})$ (i.e. M as subject) Note: In alternative method, PoT error forgetting km→m conversion gives 6.46×10^{14} (kg) for 2 marks. <u>Examiner's Comments</u> Many candidates successfully used the equation for Kepler's Third Law, which is encouraging. A quicker route was to find the Phobos's acceleration (from v^2/r) and equating that to the gravitational field strength at Phobos from Mars (GM_{mars}/r^2) and then rearranging to find the mass of Mars.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			Total	6	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
29	a		$V_{(E)} = -\frac{GM}{r}$	B1	Examiner's Comments The expression for gravitational potential was listed on the Data, Formulae and Relationships in the module 5 section and was hence reproduced well by the candidates. The minus sign was required.
	b	i	KE = $\frac{1}{2}mv^2$ and GPE = GMm/r $\frac{1}{2}mv^2 = GMm/r$ then a valid step to $v = \sqrt{(2GM/r)}$	C1 A1	Allow $m = 1$ (kg) if clearly defined Examiner's Comments Examiners were delighted that candidates proved the relationship for escape velocity very clearly indeed with the higher ability candidates correctly suggesting that 'KE + GPE = 0' was the condition for escape, although 'KE lost = GPE gained' would have been a clear way of reconciling any minus sign confusion. A minority of candidates tried, unsuccessfully, to invoke the expression for circular motion inappropriately.
		ii	$(v^2 = 2 \times 6.67 \times 10^{-11} \times 0.131 \times 10^{23} / 1.19 \times 10^6)$ $v = 1200 \text{ (m s}^{-1}\text{)}$	A1	Answer to 3.s.f. is 1210 Examiner's Comments Approximately four-fifths of all candidates calculated the escape velocity on Pluto correctly. Those that did not score the mark for this item did so because of improper calculator use or, more rarely, because they selected the wrong data from the question.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	Mercury has a higher escape velocity than Pluto (ORA) Mercury is closer to sun and Mercury is <u>hotter</u> (ORA) Molecules on Mercury (are more likely to) have speed higher than the escape velocity	B1 M1 A1	Allow a supporting calculation (speed is about 4.2 km s^{-1}) Allow 'required speed' for 'escape velocity' Allow 'fast enough to escape' Examiner's Comments Candidates found this last item very challenging indeed, with only exceptional candidates gaining two or three marks. Many candidates suggested that the reason for Mercury's lack of atmosphere was the superior gravitational pull of the Sun, which is wholly incorrect. Others suggested that the solar wind or 'radiation' had burnt off the atmosphere. Rather fewer candidates correctly related Mercury's smaller mean distance to the Sun and its higher temperature or reasoned that Mercury's escape velocity was higher than Pluto's. Only a small minority of candidates recognised that even though Mercury has a higher escape velocity, its higher temperature gave the atmosphere's molecules a higher average speed which would have exceeded Mercury's escape velocity.
			Total	7	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
30		i	Correct substitution of $T = 2.0$ s into $T^2 = \frac{4\pi^2}{g} L$ length = 0.99 (m)	C1 A1	Note: 1 (m) here cannot score this A1 mark Examiner's Comments A large majority of candidates successfully showed that the pendulum length should be 0.99m for a 'tick' length of 1.0 seconds. Candidates that attempted the reverse argument, by assuming a length of 1 m and then calculating the corresponding length, were usually unable to show the period of the resulting pendulum was 2.01s. Candidates that showed how to arrive at this period gained full credit.
		ii	Lower g / gravitational field strength / acceleration (of free fall) on Moon. T is longer (on Moon) and justified by $T^2 = \frac{4\pi^2}{g} L$ or $T^2 \propto 1/g$ or $\frac{4\pi^2}{g}$ is larger	B1 B1	Accept 'g is a sixth of g on Earth' AW Not gravity (is less) Examiner's Comments Many candidates suggested that g is less on the Moon than it is on the Earth, gaining one mark of credit. Most candidates suggested that would mean the period of the pendulum would be larger, but did so without justification from the formula in the question or contradicted themselves by stating that would make the pendulum 'run faster'.
			Total	4	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
31	a		Ensure largest possible proportion of flask is immersed. Make volume of tubing small compared to volume of flask. Remove heat source and stir water to ensure water at uniform temperature throughout. Allow time for heat energy to conduct through glass to air before reading temperature.	B1 × 4	
	b	i	Pressure is caused by collisions of particles with sides.	B1	
		i	Velocity of particles (and volume of gas) are not zero at 0 °C.	B1	
		ii	1: Gradient of graph $0.75 \times 10^2 / 100 = 0.75$	C1	Alternative method Internal energy = $3/2 \times p \times V$ At $\theta = 100^\circ\text{C}$ $pV = 2.73 \times 10^2$ Internal energy = $1.5 \times 2.73 \times 10^2 = 410$ (J)
		ii	Number of moles of gas = gradient / R = $0.75 / 8.31 = 0.09$	A1	
		ii	Mass of gas = $0.09 \times 6.02 \times 10^{23} \times 4.7 \times 10^{-27} = 2.5 \times 10^{-4}$ (kg)		
		ii	2: Internal energy = $3/2 \times NkT$		
		ii	= $1.5 \times 0.09 \times 6.02 \times 10^{23} \times 1.38 \times 10^{-23} \times (100 + 273)$	C1	
		ii	= 410 (J)	A1	
			Total	10	
32		i	(Stronger) gravitational attraction between nearby galaxies affects motion / clustering of galaxies	B1	
		ii	Expansion rate may not have been constant / non-linear expansion / effect of dark energy causing accelerating rate of expansion	B1	
			Total	2	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
33	a		Labelled diagram showing a line joining a planet and the Sun	B1	
			Comparing swept areas at different parts of orbit	B1	
	b		Arrow acting along line from planet towards sun	B1	Any arrow length
	c		Appropriate test proposed, e.g. $T^2 / r^3 = \text{constant } k$	B1	
			Test carried out on all three pairs of data	M1	$k = (1.112, 1.109, 1.113) \times 10^{-5}$ respectively
			Conclusion consistent with test result	B1	
			Total	6	
34	a		Force is proportional to the product of the mass of each asteroid. and the force is inversely proportional to the distance squared between the centres of mass of the asteroids.	B1	
	b	i	Any sensible suggestion, e.g. Satellites used for global communication, instant access to news, weather forecasting etc.	B1	
		ii	$g = (6400/15300)^2 \times 9.81$	C1	
		ii	$g = 1.72 \text{ (N kg}^{-1}\text{)}$	A1	
		iii	Acceleration towards centre = 1.72 ms^{-2} or centripetal force = mass of satellite $\times 1.72 \text{ N}$	C1	ecf (b)(i)
		iii	$T^2 = 4 \times \pi^2 \times 1.53 \times 10^7 / 1.72$	C1	
		iii	$T = 1.87 \times 10^4 \text{ (s)}$	A1	Allow 1.9
	c		Use of $M = gr^2 / G$ (accept any subject)	C1	Calculation using $g = 1.72$ at radius of 15300 km Possible ecf from (b)(i)
			Density = $3g / 4\pi G = 3 \times 9.81 / 4\pi \times 6.4 \times 10^6 \times 6.67 \times 10^{-11}$	C1	Density = $\frac{3 \times 1.72 \times (1.53 \times 10^7)^2}{4\pi \times (6.4 \times 10^6)^3 \times 6.67 \times 10^{-11}}$ $= 5.50 \times 10^3 \text{ kg m}^{-3}$
			$= 5.49 \times 10^3 \text{ (kg m}^{-3}\text{)}$	A1	
			Total	10	

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
35	Level 3 (5–6 marks) Clear description and correct calculations leading to value of total energy (must include the negative sign) <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Some description and some correct calculations or Correct calculations (including the negative sign) <i>There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Limited description or Limited calculations <i>The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear.</i> 0 marks No response or no response worthy of credit.	B1×6	Indicative scientific points may include: Description - Orbit above the equator / equatorial orbit - Orbit from west to east/same direction of orbit as Earth's rotation - Orbital period is 24 hours / 1 (sidereal) day / 23hrs 56mins (4 s) - Orbit is circular / above the same point on the Earth Calculation $E = (-) \frac{GMm}{r}$ $E = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times 2500}{4.22 \times 10^7} = (-) 2.4 \times 10^{10} \text{ J}$ $V = \frac{2\pi r}{T} \omega r$ $V = \frac{2\pi \times 4.22 \times 10^7}{24 \times 3600} = 3.07 \times 10^3 \text{ m s}^{-1}$ $E = \frac{1}{2}mv^2$ $E = \frac{1}{2} \times 2500 \times [3.07 \times 10^3]^2 = 1.2 \times 10^{10} \text{ J}$ - Total energy = $- 2.4 \times 10^{10} + 1.2 \times 10^{10} = - 1.2 \times 10^{10} \text{ J}$ - Allow full credit for algebraic proof using $\frac{GMm}{r^2} = \frac{mv^2}{r}$, $E = (-) \frac{GMm}{r}$, $E = \frac{1}{2}mv^2$ and total energy = KE + PE Allow higher order answers in terms of Lagrange's Identity Examiner's Comments This part explored multiple ideas about geostationary orbits. It was accessible to most candidates, many of whom calculated the magnitude of the GPE correctly yet forgot that this value must be negative. Almost all candidates forgot that Gravitational Potential Energy is negative.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			Total	6	